

Ahmed Ikram | Software Engineering

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PROFILE

Final-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree, graduating June 2027. Designs and ships production systems end-to-end - service architecture, cloud infrastructure, and the CI/CD pipelines that gate every deploy - leveraging agentic coding tools (OpenCode) to build fast without trading away architectural rigor. Genuinely enjoy making software fast and cheap to run: wrote a native Rust/PyO3 extension to replace a pandas bottleneck in a live inference path and picked ARM64/Graviton over x86 for a 20–30% compute-cost cut at equal performance. Recent work spans a 10-service event-driven backend (Kafka, Redis, health-check-cascading startup), a full-stack real-time collaboration platform, and a cross-cloud (AWS + Azure) data pipeline - each shipped with 1,400+ tests and zero-static-credential, OIDC-authenticated deployments. Seeking a graduate **Software Engineering** role building systems that hold up under real load.

EDUCATION

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- On track for a First-Class Honours degree. Expected Graduation: June 2027

High School of Dundee

Aug 2017 - June 2023

- Advanced Higher Computer Science (C), Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)

TECHNICAL SKILLS

Languages: Python · TypeScript · JavaScript · SQL · Bash · Rust

Frontend & Mobile: React · React Native · Expo · Tailwind CSS · Vite · Socket.IO

Backend & APIs: Flask · FastAPI · Node.js · REST API Design · Kafka · Gunicorn · Airflow · dbt · Pydantic

System Design: Microservices · Event-Driven Architecture · Distributed Systems · Idempotent Consumers · Caching Strategies · Rate Limiting · Load Balancing · Fault-Tolerant Design

Security: JWT · OAuth 2.0 · RBAC · bcrypt · AWS Secrets Manager · Encryption at Rest

Databases: PostgreSQL · Redis · SQLite · MySQL · MongoDB · Alembic

Cloud & DevOps: AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, WAF, IAM, CloudFront, SageMaker, Bedrock, Secrets Manager) · Azure (Databricks, Azure SQL, ADLS) · Terraform · Docker · GitHub Actions CI/CD · OIDC Federation

Observability & Testing: Grafana · Prometheus · StatsD · Pytest · Vitest · Jest · Playwright · Cypress · React Testing Library · dbt Tests · Mutation Testing (mutmut) · Property-Based Testing (Hypothesis)

PROJECTS

DevSync Project Management Platform with GitHub Integration

React · Tailwind · Python · Flask · PostgreSQL · Socket.IO · GitHub OAuth · AWS · Docker · GitHub Actions CI/CD

- Full-stack collaboration platform - Flask REST API over a 12-table indexed PostgreSQL schema, and a JWT-authenticated Socket.IO layer with project-scoped rooms broadcasting live task updates, comments, and dashboard refreshes with zero cross-project leakage; chose Gunicorn's gevent worker over uWSGI/ASGI so WebSocket and HTTP traffic share a single async-I/O process without async-ifying the whole codebase
- Three-tier RBAC enforced via route decorators - an undecorated route is intentionally public, not accidentally so; GitHub OAuth 2.0 handled entirely server-side with state-parameter CSRF validation and tokens encrypted at rest, never reaching the browser; bidirectional Issue/PR ↔ task sync keeps DevSync and GitHub state consistent in both directions
- AWS deployment on a custom three-tier VPC (Internet → ALB → ECS → Fargate → private RDS, no public database) with OIDC-federated GitHub Actions (zero static credentials) and health checks, eliminating API/UI version mismatch
- Multi-stage Docker build cut the backend runtime image 45% (600MB → 330MB) by isolating build tooling (gcc, libpq-dev) to a discarded build stage; container starts (migrations + health check) in under 20s, and the API holds sub-300ms p99 latency on authenticated JSON endpoints

ATM Log Aggregation, Analysis & Diagnostics Platform

Python · FastAPI · React · PostgreSQL · Kafka · Redis · ChromaDB · nginx · Docker · Terraform · AWS

- Backend engineering lead on a 7-person Agile team delivering a diagnostics platform for NCR Atleos - a real industry stakeholder, not a solo portfolio project; independently architected a 10-service Dockerised platform including the **RAG** diagnostic assistant, full AWS production deployment, and 3-pipeline CI/CD system - all built post-deadline on top of the team-delivered foundation
- 10-service system with health-check cascading startup: FastAPI REST API (30 endpoints, 6 routers, JWT + Redis blacklist auth), Kafka Kraft consumer with at-least-once delivery with idempotent deduplication (hybrid Redis SET+TTL primary, 10K-entry LRU fallback) and a Redis Stream dead-letter queue, 8 Redis distributed patterns, and a React frontend shipped via a multi-stage Docker build (~25MB nginx-alpine final image)
- Deployed on AWS via Terraform (10 modules, 118 resources): multi-AZ VPC (2 public + 2 private subnets), ECS Fargate (2 services × 2 tasks across AZs) behind an ALB, an EC2 host running Kafka/Redis/ChromaDB/Ollama in a private subnet, CloudFront serving the React SPA from S3, 8 secrets injected at container start, and 6 least-privilege IAM roles
- 3-pipeline CI/CD system: CI runs ruff/mypy/bandit lint, Checkov IaC compliance, and 1,402 tests (923 pytest, 394 vitest, 10 Playwright E2E, 75 Terraform assertions) in parallel on every PR; a path-filtered CD pipeline skips infra jobs on docs-only changes and triggers a zero-downtime ECS rolling update on merge to main - all authenticated to AWS via OIDC, zero long-lived credentials

StockLens FinTech Spending & Investment App

React Native · Python · FastAPI · PostgreSQL · Redis · Rust · Terraform · Docker · AWS

- Async FastAPI backend exposing 59 REST API endpoints across 14 routers with biometric-backed JWT auth, bcrypt password hashing, slowapi rate limiting and tenacity-driven retry/backoff on the **LLM agent** endpoint
- Native Rust extension (PyO3/Maturin, 12 exported functions) replaces pandas-based technical-indicator coputation with zero-cost abstractions, compiled to a wheel and cross-built for ARM64 via QEMU in CI - called directly by both the LSTM training pipeline and live API with zero Python-side overhead
- Deployed on AWS via Terraform (14 modules, 193 resources): three-tier network isolation (Internet → WAF-protected ALB → ECS Fargate ARM64 → private RDS/Redis) chosen for a 20-30% compute-cost cut over x86 at equal performance, Multi-AZ RDS and TLS/AUTH-enabled ElastiCache Redis, CPU/RPS-triggered auto scaling (min=2, max=6), and zero-downtime rolling deploys gated on ECS health checks
- 1,415 backend and 822 frontend tests at a 90%/90% coverage gate, including Hypothesis-driven property-based tests on TWR/cash-flow invariants and nightly mutation testing (mutmut) on critical paths (LSTM inference, OCR parsing); enforced by a 9-parallel-job CI pipeline (lint, TypeScript, Jest, Rust clippy/cargo test, Docker/IaC/secrets scanning) and a 7-stage OIDC-authenticated CD pipeline (build → scan → plan → manual review → apply)

W3C Web Logs ETL Pipeline

Apache Airflow · Python · Databricks, dbt · Apache Spark · Azure SQL · Terraform · Docker · GitHub Actions · Power BI · Grafana · Prometheus

- Serverless Databricks medallion pipeline (Bronze → Silver → Azure SQL → dbt → Power BI) ingesting 93 IIS log files (155,380 rows, 0 dropped) across 7 '@dlt.expect_or_drop' quality gates into 153,377 GeoIP-enriched Silver rows a custom UDF handles 14- vs 18-field IIS format detection dynamically, and 2 Airflow DAGs orchestrate the full pipeline with zero data-processing logic in DAG files - layers decoupled via Airflow Datasets, no ExternalTaskSensor
- 16 dbt models (10 staging + 6 marts) implement a Kimball-style star schema across 9 dimension tables, compiling against both PostgreSQL and T-SQL from a single codebase via 18 compatibility macros; 121 dbt data tests (not_null, unique, accepted_values, relationships, expression_is_true) enforce referential integrity before data reaches Power BI
- Provisioned 30+ Azure resources across 2 Terraform configurations including the full OIDC auth chain (Azure AD app, service principal, federated credential, role assignment) via a single terraform apply; 7 GitHub Actions workflows (4 CI + CD + CodeQL + auto-merge) with 9 parallel CI job stages and a rollback workflow that re-applies Terraform from the previous commit on demand - zero static credentials anywhere
- Delivered a 7-page Power BI dashboard (156K requests, 88 countries, P95 1,149ms) from 18 dbt-generated CSV exports via a semantic contract; observability: a custom Python freshness probe refreshes 3 Grafana dashboards (23 panels) and 10 alert rules across 3 severity-tiered action groups with Slack notifications via Prometheus; 613 tests across 5 tiers including dual-dialect dbt CI against side-by-side PostgreSQL and SQL Server containers

CERTIFICATES

AWS Academy Graduate - Machine Learning Foundations

Feb 2025

AWS Academy Graduate - Cloud Foundations

Mar 2024

Foundations of Azure AI: Concepts, Capabilities, and Implementation

Oct 2024

EXPERIENCE

Beaully Mini Market, Dundee, Scotland

Retail Assistant

September 2023 - Present

- Troubleshoot and resolved Point-of-Sale and stock system errors quickly to minimise customer wait time

Zulus Peri Peri Grill, Dundee, Scotland

Front of House Worker

June 2021 - September 2023

- Trained 5 new staff members, helping them get up to speed quickly during busy service periods